

```
1 using CustomProjectRPG.Interfaces;
2 using System;
3 using System.Collections.Generic;
4 using System.Linq;
5 using System.Text;
6 using System.Text.Json;
7 using System.Threading.Tasks;
8 using Enum = CustomProjectRPG.Entities.Enum.Enum;
9 using System.Text.Json.Serialization;
10 using CustomProjectRPG.Core.Utilities;
11
12
13 namespace CustomProjectRPG.Entities.Component
14 {
15
16
17     public class StatsComponent : ISerializeJSON<StatsComponent>
18     {
19         private float _hp;
20         private float _maxHp;
21         private float _atk;
22         private float _def;
23         private float _agility;
24         private Enum.Enum.EntityType _entityType;
25
26         public float HP => _hp;
27         public float MaxHP => _maxHp;
28         public float ATK => _atk;
29         public float DEF => _def;
30         public float Agility => _agility;
31
32
33         /// Initializes a new StatsComponent with base stats for a           ↗
34         /// specific entity type.
35
36         public StatsComponent(Enum.Enum.EntityType entityType, float           ↗
37             initialHp = 100f, float initialAtk = 10f, float initialDef =           ↗
38             5f, float initialAgility = 5f)
39         {
40             _entityType = entityType;
41             _maxHp = initialHp;
42             _hp = _maxHp;
43             _atk = initialAtk;
44             _def = initialDef;
45             _agility = initialAgility;
46         }
47
48         /// Applies damage to the entity's HP, ensuring it doesn't go           ↗
```

```
        below 0.

47
48     public void TakeDamage(float amount)
49     {
50         if (amount < 0) throw new ArgumentException("Damage must be  ↗
            non-negative.", nameof(amount));
51         _hp = Math.Max(_hp - amount, 0f);
52     }
53
54
55     /// Heals the entity, capping HP at MaxHP.
56
57     public void Heal(float amount)
58     {
59         if (amount < 0) throw new ArgumentException("Heal amount must  ↗
            be non-negative.", nameof(amount));
60         _hp = Math.Min(_hp + amount, _maxHp);
61     }
62
63
64     /// Applies corruption-based stat penalties using the provided  ↗
        CorruptionComponent.
65
66     public void ApplyCorruptionPenalty(ICorruption corruption)
67     {
68         if (corruption == null) throw new ArgumentNullException(nameof( ↗
            corruption));
69         if (_entityType != Enum.Enum.EntityType.Player) return;
70         float multiplier = corruption.GetStatMultiplier();
71         _atk *= multiplier;
72         _def *= multiplier;
73         _agility *= multiplier;
74     }
75
76
77     /// Resets stats to their base values (e.g., after a loop reset).
78
79     public void ResetStats()
80     {
81         _hp = _maxHp;
82         _atk = 10f;
83         _def = 6f;
84         _agility = 5f;
85     }
86
87
88     /// Serializes the stats state to JSON.
89
90     public string Serialize()
```

```
91     {
92         var data = new StatsData
93         {
94             HP = _hp,
95             MaxHP = _maxHp,
96             ATK = _atk,
97             DEF = _def,
98             Agility = _agility
99         };
100     return SerializeJSON.ToJson(data);
101     }
102
103
104     /// Deserializes the stats state from JSON.
105
106     public void Deserialize(string json)
107     {
108         var data = SerializeJSON.FromJson<StatsData>(json);
109         _hp = Math.Clamp(data.HP, 0f, data.MaxHP);
110         _maxHp = data.MaxHP;
111         _atk = data.ATK;
112         _def = data.DEF;
113         _agility = data.Agility;
114     }
115
116     private class StatsData
117     {
118         [JsonPropertyName("hp")]
119         public float HP { get; set; }
120
121         [JsonPropertyName("maxHp")]
122         public float MaxHP { get; set; }
123
124         [JsonPropertyName("atk")]
125         public float ATK { get; set; }
126
127         [JsonPropertyName("def")]
128         public float DEF { get; set; }
129
130         [JsonPropertyName("agility")]
131         public float Agility { get; set; }
132     }
133 }
134
135 }
136
```